FORMULAE LIST

Volume of a sphere: $V = \frac{4}{3}\pi r^3$ Volume of a cone: $V = \frac{1}{3}\pi r^2 h$ Volume of a pyramid: $V = \frac{1}{3}Ah$

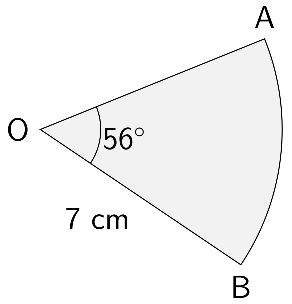
1. The 2026 *Artemis II* flight, travelling around the Moon, lasted for approximately 217 hours. (2)

   The total distance travelled was approximately 694,400 miles.

   Calculate the approximate average speed of the *Artemis II* during its flight.

   Give your answer in scientific notation.

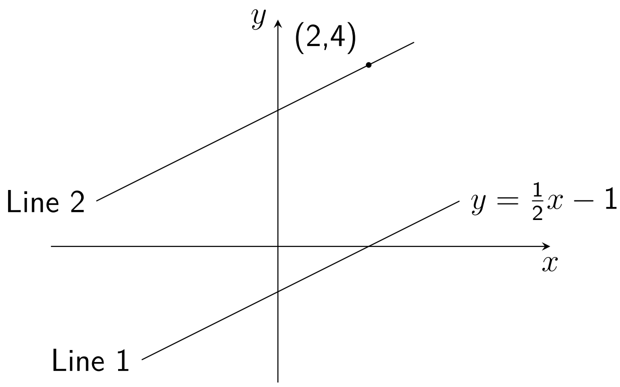
2. The diagram below shows a sector of a circle, centre O. Angle AOB is 56°. (3)



   Calculate the area of the sector OAB.

   Give your answer correct to three significant figures.

3. The two lines shown in the diagram below are **parallel**. (3)



   Determine the equation of Line 2.

4. The volume of water in a pond is expected to decrease by 3.2% each day during the summer. (3)

   Today the volume of water is measured as 4800 litres.

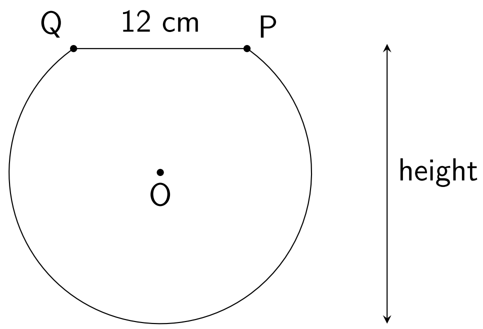
   Calculate the expected volume of water in the pond **one week** from today.

5. Simplify $\dfrac{(2x^5)^3}{4x^8}$. (3)

6. Determine the coordinates of the point where the straight line with equation $2x + 3y + 5 = 0$ intercepts the y-axis. (2)

7. A shape is in the form of part of a circle with centre O, as shown below. (4)

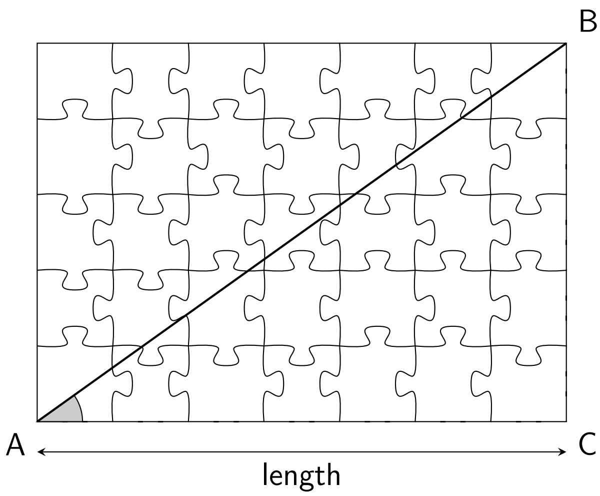
   The circle has radius 11 centimetres, and distance PQ is 12 centimetres.



   Determine the height of the shape.

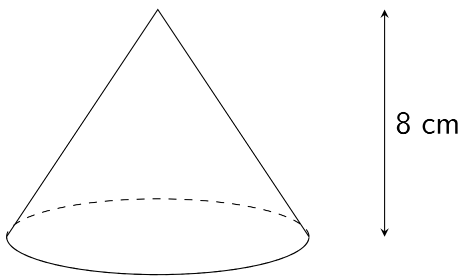
8. A rectangular jigsaw puzzle has a diagonal of 8.2 centimetres, shown as line AB below. (3)

   The shaded angle BAC is 36°.



   Calculate distance AC, the length of the jigsaw.

9. The cone shown below has a height of 8 centimetres and a volume of 230 cubic centimetres. (4)



   Calculate the radius of the cone. **Give your answer correct to three significant figures.**

END OF PRACTICE EXAM B PAPER 2

HINTS: Practice Exam B Paper 2

1. The formula to calculate average speed is: speed $= \dfrac{\text{distance}}{\text{time}}$
2. The area of a sector is a **fraction** of the **area of a circle**. What is the **formula**?
3. What is the gradient of Line 1? Once you have the gradient of a line and any point it passes through, you can use the equation $y - b = m(x - a)$
4. What is 100% minus 3.2%? How many days are in a **week**?
5. Remember the three **index laws**, and remember that the power of 3 also applies to the **coefficient** in $2x^5$.
6. Rearrange the equation of the straight line to **make y the subject**.
7. Aim to isolate (and sketch) a **right-angled triangle**, before using Pythagoras' Theorem. Draw a line vertically from the centre of the circle to meet the line PQ.
8. Sketch a **right-angled triangle** and *label* the **opposite**, **adjacent** and **hypotenuse**.
9. Substitute the *volume* and **height** into the **formula for the volume of a cone** and then **solve**.

ANSWERS: Practice Exam B Paper 2

1. 3.2×10^3 miles per hour
2. 23.9cm^2
3. $y = \frac{1}{2}x + 3$ **or** $2y = x + 6$
4. 3822.7 litres
5. $2x^7$
6. $(0, -\frac{5}{3})$
7. 20.2 cm
8. 6.63 cm
9. 5.24 cm